MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL
Paper Code: PCC-CSBS 502 Database Management Systems UPID: 005963

Time Allotted: 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following: [1 x 10 = 10]

(I) Write the significance of keys in the B-Tree.
(II) What is Serializability in transaction?
(III) What is the derived attribute in ER model?
(IV) A relation is in 2NF if it is in 1NF and all its non-key attributes are ________.
(V) "Candidate key is a minimal subset of a super key"-justify this statement.
(VI) The result of which join operation contains all pairs of tuples from the two relations, regardless of whether their attribute values match.
(VII) What is the difference between clustered and Non-clustered indexes in the database?
(VIII) How rigorous 2PL protocol is better than a strict 2PL protocol?
(IX) Write the main roles of a database administrator.
(X) Write two main differences between OLTP and OLAP processes.
(XI) How can we avoid update anomalies in DBMS?
(XII) What is the difference between indexing and hashing in dbms?

Group-B (Short Answer Type Question)

Answer *any three* of the following: [5 x 3 = 15]

2. Employee schema: Emp={emp#, name, deptname, manager, age} [5]
   (I) Write a query in relational algebra to find emp#, name, and age working under production and HR departments both.
   (II) Write a query in relational algebra to find the name and age working under the manager ‘Robin’.
3. Consider the employee database:
   working (empid, dept_name, emp_name, emp_address) salary [5]
   (empid, empsal)
   Write the appropriate query in SQL on the basis of the above schema.
   i. Find the salary of employees whose department is the same as the department of ‘Amrita’.
   ii. Find the average salary of employees where the employee name ended with ‘it’.
4. Explain some common threats and challenges in databases. [5]

5. Consider a relation scheme R = (A, B, C, D, E, F, G) on which the following functional dependencies hold: {A–>B, BC–>D, BC->E, AEF–>G, B–>G}. What are the candidate keys and super keys of R? Justify with the explanation. [5]

6. What is the necessity of using the normalization technique in RDBMS? Explain all the types of anomalies with suitable examples. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following: [15 x 3 = 45]

7. (a) State two advantages of DBMS over traditional file systems. [5]

(b) Develop an extended ERD for an online Ticket Reservation system (show all possible enhanced features, Generalization/Specialization, Disjoint/Overlapping constraints, Total/Partial constraints, cardinality, and Aggregation). [10]

8. (a) Briefly explain the role of query optimizer with respect to DBMS. Define union and intersection compatibility with example. [5]

(b) Differentiate between Cartesian product and Natural join through relational algebra. [10]

9. (a) Discuss the various properties of B-Tree. [5]

(b) Construct a B-Tree of Order 3 by inserting numbers from 1 to 10. Explain each step of the operation clearly. [10]

10. (a) Explain different types of transaction states with a diagram. [5]

(b) Check whether the given schedule S is conflict and view serializable or not- [10]

- S1: R1(X) R1(Y) R2(X) R2(Y) W2(Y) W1(X)
- S2: R1(X) R2(X) R2(Y) W2(Y) R1(Y) W1(X)

11. (a)

X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

a) Which of the functional dependency holds good for the above-given data? Answer with a proper explanation. [4]

XY-->Z && Z--> Y
YZ-->X && Y--> Z
YZ-->X && X-->Z
XZ-->Y && Y-->Z
Why the others do not hold?

(b) Given R (A, B, C, D, E) with the set of FDs, F= {A→BC, CD→E, B→D, E→A}
(i) Find the candidate keys of R. [5]

(c) What are the various users and their roles in DBMS? [6]

*** END OF PAPER ***